

Candidate's Name:

School:

Centre No.				Personal No.		
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Sign:

553/2
BIOLOGY
PRACTICAL
Paper 2
July/August 2023
2 hours



HOIMA DIOCESE EXAMINATIONS BOARD

UCE Mock Examination, 2023

BIOLOGY (PRACTICAL)

Paper 2

2 hours

INSTRUCTIONS TO CANDIDATES:

This paper consists of three questions.

Answer all questions.

Drawings should be made in spaces provided.

Use sharp pencils for your drawings.

Coloured pencils or crayons should not be used.

No additional sheets of writing paper are to be inserted in this booklet.

For Examiners' Use Only

QUESTION	MARKS	EXAMINER'S SIGNATURE AND NUMBER
1.		
2.		
3.		
TOTAL		

1. You are provided with solution *X* which contains a food nutrient and chemical reagent *Z*. Label test tubes *A* to *E*. Label another test tube *Z*.
- (a) Using a measuring cylinder measure the following quantities of solution *X* into test tubes labeled *A* to *E*.

Table 1

Test tube	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>
Volume of <i>X</i> (cm ³)	1	2	3	4	5

- (i) Increase the total volume of solutions in the test tubes *A*, *B*, *C*, *D* and *E* to uniform volume of 5 cm³ by adding distilled water.
- (ii) Record the volume of distilled water added to each test tube in **Table 2** below. (02½ marks)

Table 2

Test tube	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>
Volume of water (cm ³)					

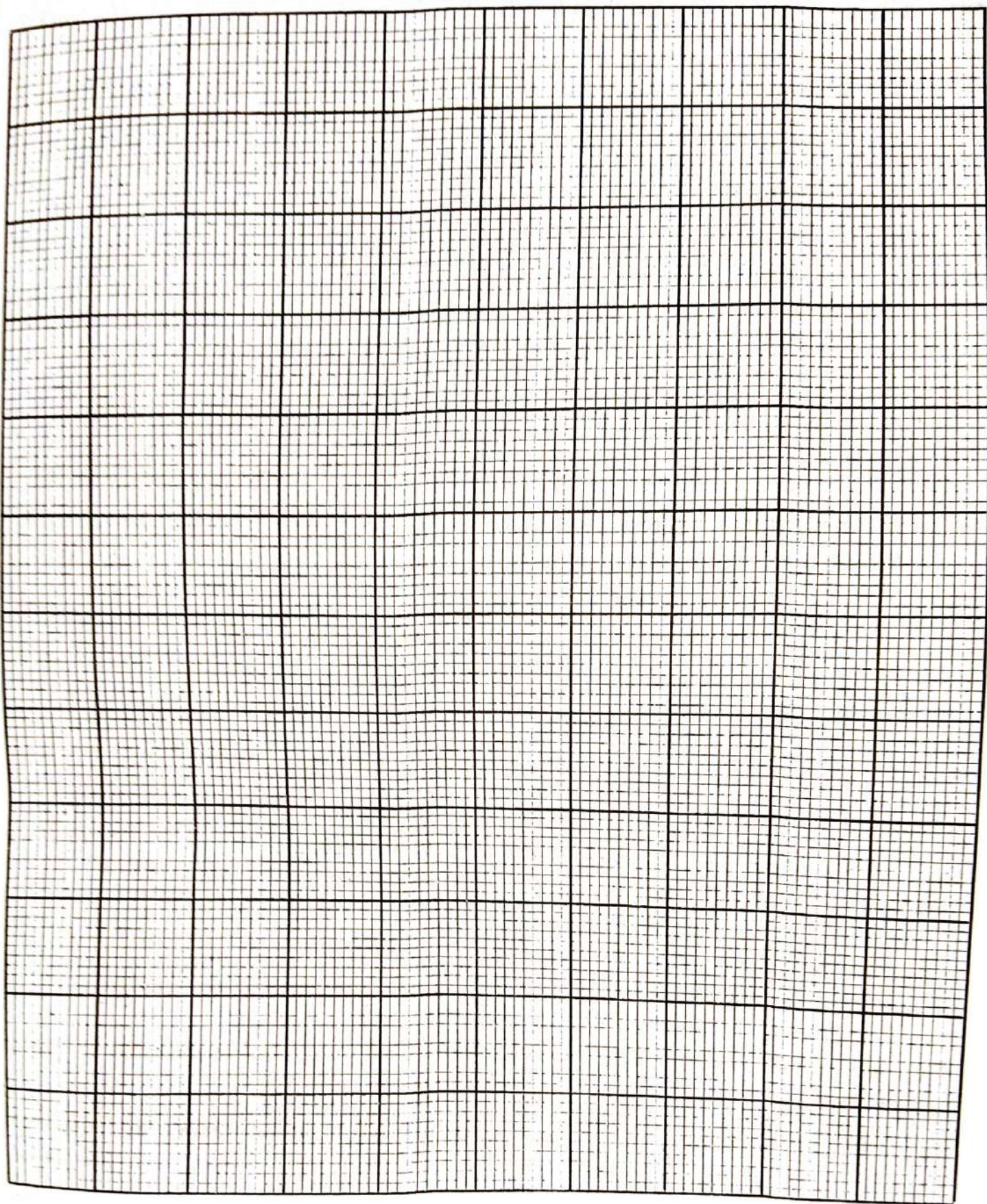
- (iii) Measure 1 cm³ of solution *Z* into test tube *Z*. Using a dropper, add the solution from test tube *A* dropwise to solution *Z* in test tube *Z* while shaking the test tube. Count the number of drops required to decolourise solution *Z* and record your results in **Table 3** below. Pour the mixture in test tube *Z* and wash it thoroughly.

Repeat the procedure above using solution from test tubes *B*, *C*, *D* and *E*. (02½ marks)

Table 3

Test tube	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>
Number of drops required to decolorize solution <i>Z</i>					

- (b) On the grid below, represent the relationship between the number of drops added (from **Table 3**) and volume of *X* (from **Table 1**). (06 marks)



(c) Explain the effect of addition of water.

(02 marks)

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(d) Giving a reason, suggest the identity of:

(04 marks)

- (i) Food nutrient in solution **X**.
Identity

Reason

- (ii) Solution **Z**.
Identity

Reason

- (e) In which of the test tubes **A** to **E** do we find the solution that would be best useful to someone with bleeding gums? Give a reason for your answer. (02 marks)

- (f) What conclusion can be made from the results in (a) above? (01 mark)

2. You are provided with specimen **G**. Examine it carefully and answer the following questions.

- (a) Giving two reasons in each case, state the **phylum** and the **class** of the specimen. (05 marks)

- (i) Phylum:

Reasons

(ii) Class

Reasons

- (b) Hold the specimen with **one** lateral side uppermost. Slowly and carefully, slide one finger from the head to the tail. State what is observed and give the significance of the above observation to the specimen. (02 marks)

Observation.

Significance of the above observation.

- (c) Remove one scale from the lateral side. Try to tear it. (02 marks)

(i) State what is observed.

(ii) State the significance of the above observation to the specimen.

- (d) State **three** adaptations of specimen **G** for survival in its habitat. (03 marks)

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- (e) Describe the **position** and **structure** of the dorsal fin of specimen **G**. (03 marks)

Position

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Structure

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- (f) Cut off the trunk and head of specimen **G**. Draw and label the remaining part. (05 marks)

3. (a) Examine the shoot of specimen *K* and:

(04 marks)

(i) Describe the leaves.

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(ii) State the class of specimen *K*. Give one reason for your answer. (02 marks)

Class

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Reason

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(iii) Explain how the leaves adapt specimen *K* for survival in its habitat. (03 marks)

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(b) Examine the root system of specimen *K*.

(i) Count and record the number of secondary roots (lateral roots) and the number of tertiary roots on the longest secondary root in **Table 4** below. (02 marks)

Turn Over

Table 4

Number of secondary roots (lateral roots)	
Number of tertiary roots.	

- (ii) Basing on your results in **Table 4**, explain how the roots of specimen **K** adapt it for survival in the terrestrial environment. (03 marks)

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- (c) Now pluck off one mature leaf from specimen **K** ensuring that none of its parts remains attached on the stem. Cut of the lamina of the leaf. Draw and label the underside of the remaining part. (06 marks)

END